

06. THE 3 EMBRYOLOGICAL TISSUES

DURATION : 47:51

STUDY SHEET

COURSE SUMMARY

This session explores the three embryological tissues: the ectoderm, endoderm, and mesoderm, linking them to biodynamic concepts. By introducing Blech-Schmidt's classification, the teaching highlights the relationship between boundary and interior tissues, illustrated by concrete examples such as the eardrum. The importance of connective tissues, their nutritive role for epithelial tissues, and the dynamics of congestion and desiccation are discussed. Understanding these elements will enable you to integrate a holistic approach into your therapeutic practice.

THE 3 EMBRYOLOGICAL TISSUES: A BIODYNAMIC APPROACH

EMBRYONIC AND HISTOLOGICAL TISSUES: BEYOND THE CLASSICS

In classical embryology, we distinguish three fundamental tissues: **ectoderm**, **endoderm**, and **mesoderm**. In histology, this classification is simplified, and we can summarise them into two main types of tissues:

- **Epithelial tissues:**

- * Of ectodermal origin (linked to the nervous system).

- * Of endodermal origin (linked to the digestive system).

- **Connective tissues:**

- * Of mesodermal origin.

BLECH-SCHMIDT'S VISION: BOUNDARY TISSUES AND INTERIOR TISSUES

Blech-Schmidt introduces an interesting perspective by classifying tissues according to their function and position:

- **Boundary tissues:** Ectoderm and endoderm.
- **Interior tissues (or connective tissues):** Mesoderm.

To facilitate understanding, we will use a colour code:

- **Red:** Endoderm (digestive system).
- **Blue:** Ectoderm.
- **Green:** Mesoderm.

This recognition allows us to orient ourselves within the system and perceive its dynamics.

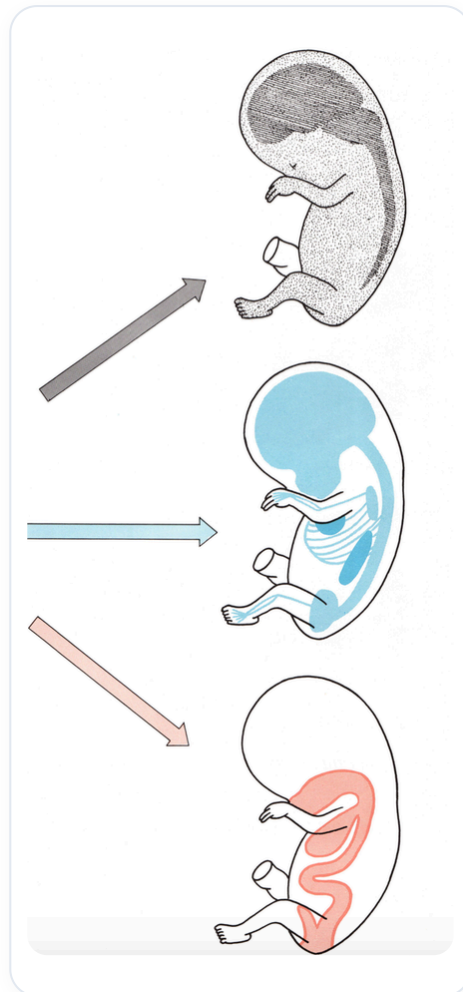


FIG. — Embryonic germ layers

THE LINK BETWEEN HISTOLOGY, EMBRYOLOGY, AND BIODYNAMICS

This approach builds a bridge between histology, classical embryology, and biodynamic embryology.

CHARACTERISTICS OF EPITHELIAL (ECTODERMAL) TISSUES

An epithelial tissue, like the ectoderm, is a tissue concentrated between a fluid and an interior tissue. Similarly, an endodermal tissue is often concentrated near the body's internal "seas".

THE EARDRUM: A CONCRETE EXAMPLE

The eardrum is a perfect illustration of this organisation:

- **Outside:** Ectodermal tissue.

- **Inside:** Endodermal tissue.
- **In the middle:** Mesodermal tissue.

THE DEPENDENCE OF BOUNDARY TISSUES

Boundary tissue (epithelial) is always dependent on interior tissue (connective). Vessels and nutrition come from the interior tissue and are distributed to the boundary tissue. In other words, mesodermal tissue contains the trophic information necessary for the growth and maintenance of other tissues.

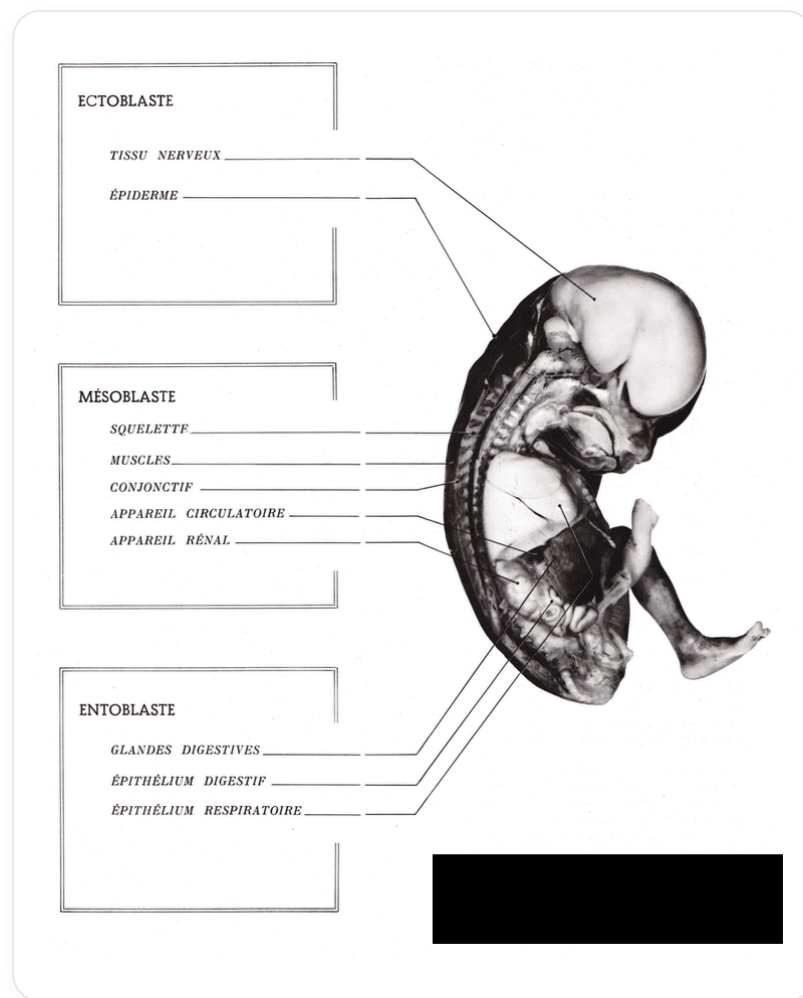


FIG. — Embryonic germ layers

TISSUE CONGESTION AND DESICCATION

There is no congestion in boundary tissue, but phases of congestion can occur in interior tissue.

Imagine a desiccated area: it needs to be rehydrated, reopened, reconnected. This can be congestion or a phase of desiccation. These phenomena occur in life: dryness makes the tissue dry, while excess water can create congestion.

Women experience natural congestions every month, such as uterine congestion, which is a cleansing process in preparation for welcoming life.

THE ROLE OF MESODERMAL TISSUE

Mesodermal tissue is the body's **ground**. It gives rise to:

- The entire fluid system (vessels, arteries, veins, lymphatic system).
- The entire skeletal and musculoskeletal system (bones, membranes, peritonea).
- The circulatory system.
- The urogenital system.
- The entire support system.

It is the quintessential interior tissue.

CHARACTERISTICS OF BOUNDARY AND INTERIOR TISSUES

- **Boundary tissue:**

- * Great variety of forms.
- * Surface growth.
- * No congestion phase.
- * Many cells.
- * Always dependent on interior tissue.

- **Interior tissue:**

- * Located between boundary tissues.
- * Can undergo congestion.

If the interior tissue is damaged, it can create a field of corrosion.

THE COMPOSITION OF CONNECTIVE TISSUE

Connective tissue is composed of four main elements. If we compare a bone to a tendon:

- **Bone:** Many cells, little water, many minerals.
- **Tendon:** Fewer cells, more water, and a different matrix with more elastin.

GLYCOSAMINOGLYCANS AND TISSUE DENSITY

Glycosaminoglycans play a crucial role in matrix structure. We observe two types of structures:

- **Permeable (depolymerised):** More fluid.
- **Dense (polymerised):** Harder, like ligaments and tendons.

All mesodermal structures can be either permeable or viscous. The more fluid they are, the fewer constraints there are.

THE SPHENOBASILAR SYNCHONDROSIS: THE BODY'S ZERO POINT

Observing the teeth and occlusion gives us a perfect picture of the constraints at the sphenobasilar synchondrosis. This synchondrosis is an embryological key for the body, a true **zero point** that receives all information:

- Vascular
- Lymphatic
- Neurological
- Digestive
- Tension membranes

The sphenobasilar synchondrosis is aware of everything that happens. The teeth adapt to maintain freedom at this level.

TEETH AND VERTEBRAL AXIS

There is an important relationship between the teeth and the notochord, or more precisely, the vertebral axis. Teeth can disrupt static posture, and a slipped disc can be treated by balancing occlusion.

HEALTH: A DYNAMIC BALANCE

Health is a balance, a harmony that expresses a state of well-being. Let's take the example of a tendon:

- **Healthy tendon:** Free, dynamic, in balance between constraint and fluids.
- **Imbalance:** Internal or external (e.g., a demanding profession). This can lead to calcific tendinitis, where the tendon hardens.

EMBRYONIC LINKS AND DISTANT EXPRESSIONS

Limbs are an expansion of a peritoneal sac in a specific movement. A liver or lung problem can block the rotation system and express itself distantly, for example, at the elbow or wrist.

HOMEOSTASIS AND MAGNETIC FORCE

Homeostasis is the set of processes that allow the body to maintain its balance. For this, it needs a **magnetic force**, electricity, a voltage.

THE MICRO-CRYSTALLINE FIELD AND TOUCH

In biodynamic osteopathy, we work with a micro-crystalline field. The touch is extremely gentle, but global, because it acts on this field.

- The hand rests on the heart.

- The eyes rest on the heart.
- The voice rests on the heart.

The heart emits a powerful rhythmic field from the 23rd day, informing our hands and our gaze.

CRANIAL MOVEMENTS AND THEIR IMPLICATIONS

TORSION

Torsion is a movement of the sphenoid in connection with the maxilla, with the occiput and jaw remaining fixed. This can lead to facial asymmetries.

SIDE BENDING AND ROTATION

A "side bending" with rotation at the occiput has repercussions on the sphenobasilar synchondrosis and the face.

THE NOTOCHORDAL AXIS AND THE SACRUM

The osteopathic introduction highlights the notochordal axis and the development of the sacrum, as well as the primitive cranio-sacral axis. The sacrum and the base of the skull are intrinsically linked.

SUMMARY OF EMBRYONIC TISSUES

- **Ectoderm:** Boundary tissue. Corresponds to the entire nervous system and skin.
- **Endoderm:** Boundary tissue. Corresponds to the digestive and respiratory epithelium.
- **Mesoderm:** Interior tissue. Gives rise to everything skeletal, muscles, connective tissues, circulatory and urogenital systems.

THE INTERDEPENDENCE OF TISSUES

Boundary tissue is always dependent on interior tissue. We cannot talk about the development of the nervous system without its mesodermal "packaging".

THE EMBRYO: A DEVELOPMENTAL MOVEMENT

The embryo can be defined in terms of days, weeks, and size. But what is truly important is the notion of **growth** and **developmental movement**.

MOTILITY, MOBILITY, MOTOR FUNCTION

It is crucial to distinguish:

- **Motility:** An embryonic response, the movement of development itself.
- **Mobility:** A movement related to breathing.

- **Motor function:** A voluntary movement.

We do not treat an organ, but an environment. Our action can influence motility, mobility, and motor function.

THE POWER OF TISSUE AND SUBTLE PERCEPTION

A teacher once told me: "Marc, one day, you will feel the power in the tissue." By observing embryonic development, one perceives this power.

CELLULAR POLARITY: THE ECCENTRIC NUCLEUS

A fertilised egg is an egg that has received the genetic heritage of the sperm. If a cell has its nucleus in the centre, it is not polarised. In all the cells of our body, the nucleus is always eccentric.

THE SPERM ENCOUNTER AND REORGANISATION

The oocyte is polarised by small cells that create a force. When the sperm arrives, it touches the membrane, releasing a specific hormone. At the moment of encounter, there is a global reorganisation of the system.

THE BODY'S ELECTRICAL FIELD

The largest electrical field in our body is along our spine. There are also powerful electrical fields in the brain and heart.

PRACTICE: BEING A CELL AND THE STILL POINT

We will do a practice where we imagine being a cell on a table. In embryology, we start from there and we return to it.

STAYING WITH HEALTH

In biodynamic work, we recognise the lesion, but we stay with health, with its dynamics.

THE NOTION OF SPACE AND STILLNESS

An Aikido master teaches us the importance of the notion of space, unity, globality, and harmony.

RECENTRING AND NON-CONFLICT

The Aikido master shows us how to bring our consciousness and mind back to our centre of gravity.

MOVING OTHERS WITHOUT MOVING THEM

To move others without disturbing them, one must move the space in which they are located.

CONCLUSION

In biodynamic osteopathy, we work with these spaces. The body reorganises itself in relation to this space. The space is there; it is just a matter of being in alignment.
